

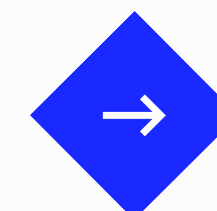


3DPR REVIEW



The Frontier of Innovation

Reviewing the 3DPR
Rocket from E1RK



01

Projected Statistics

Apogee: 57 meters

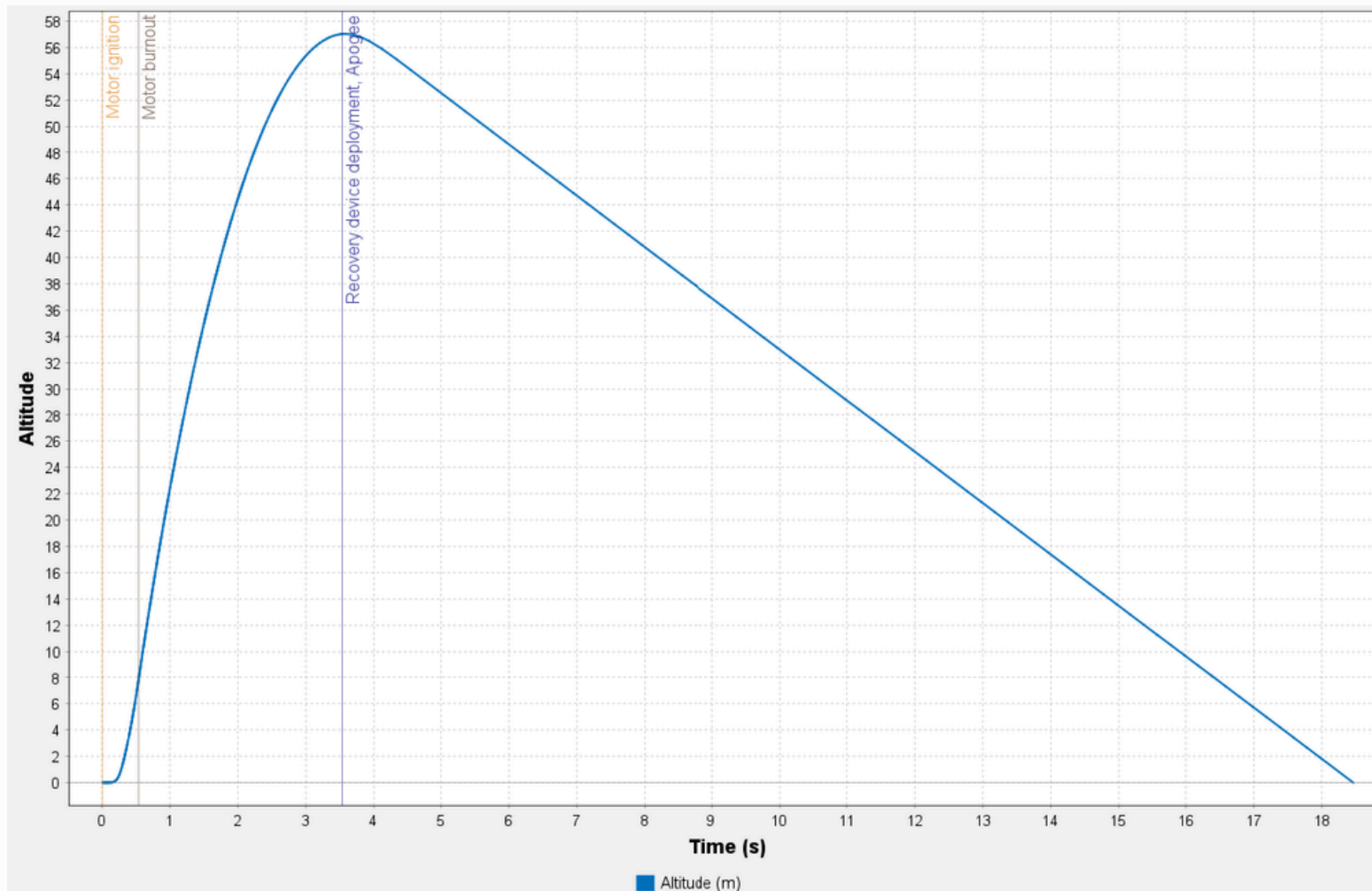
Stability: 1.8 cal

Length: 20.3cm

Predicted mass: 54.5g

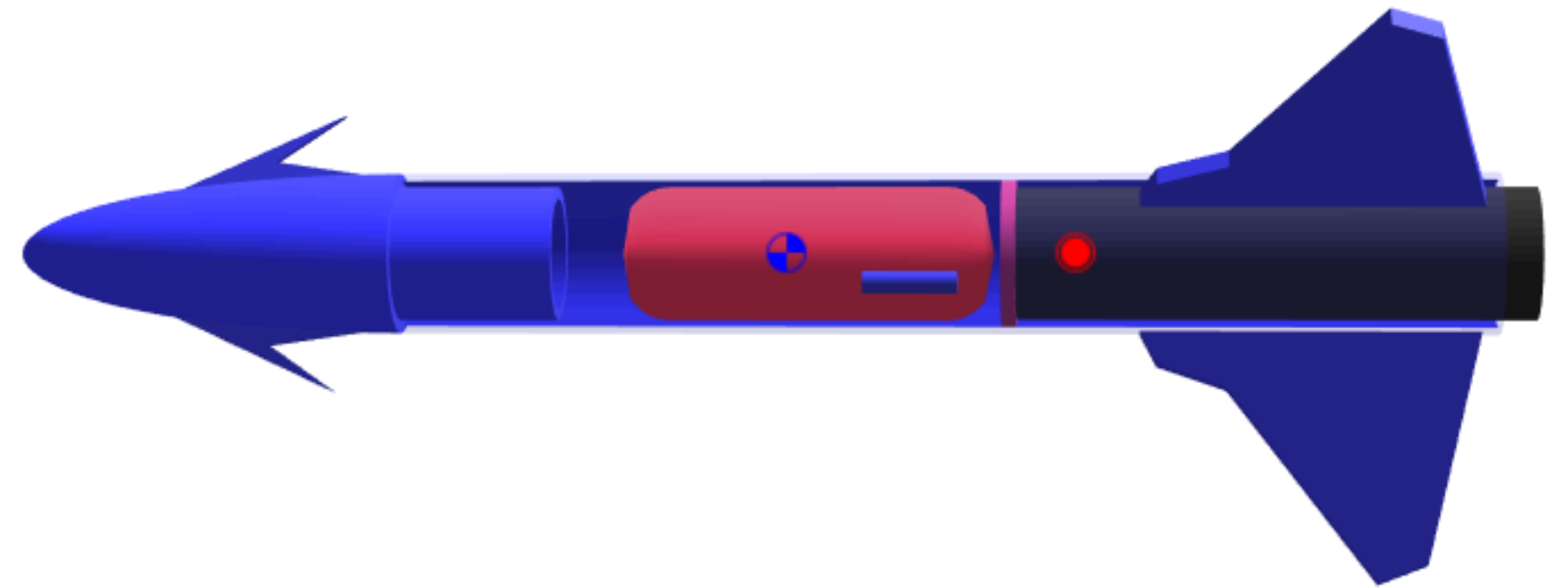
Time between apogee and
recovery deployment: 0.0 sec

TIME VS. ALTITUDE

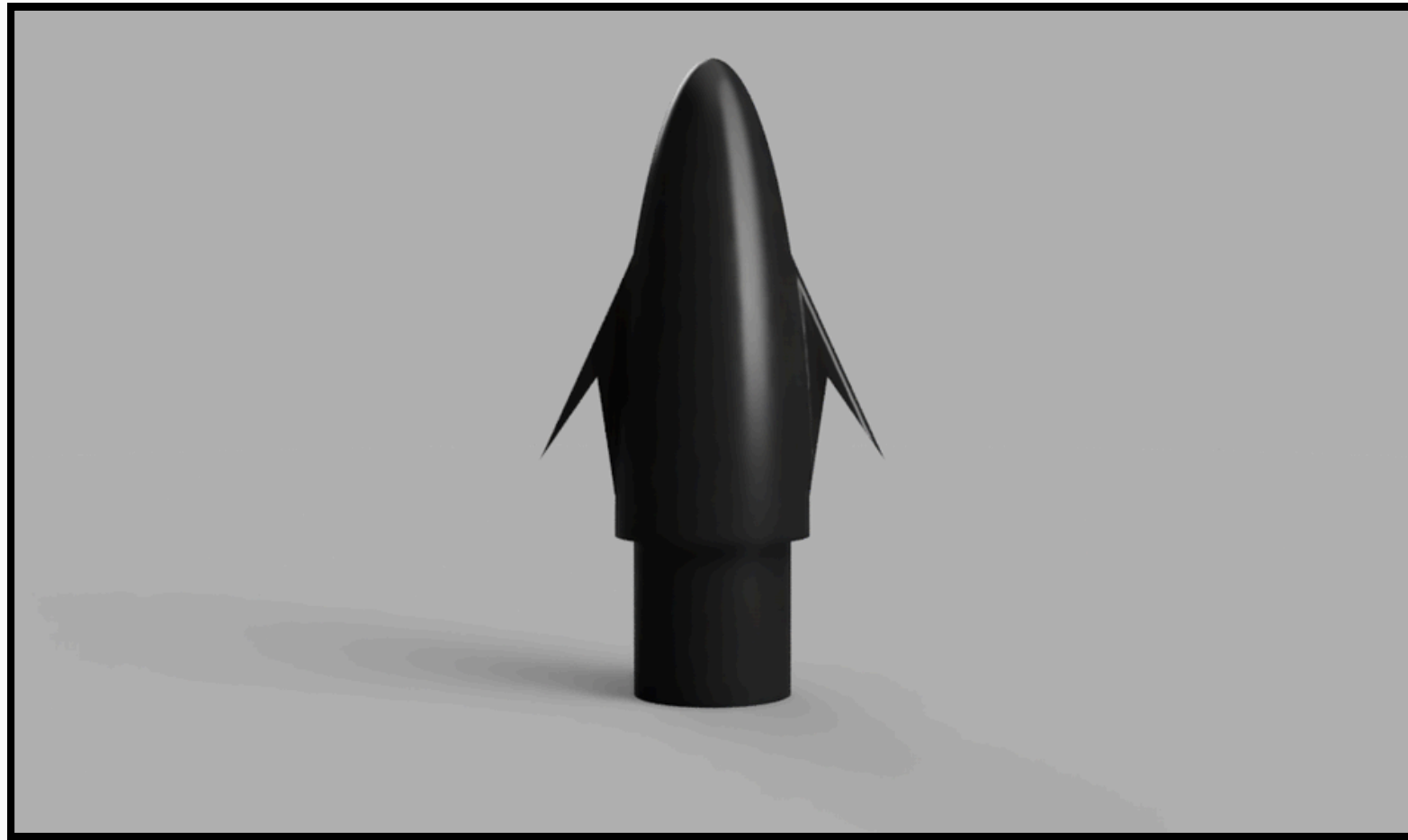


02

Empty mass: ~25g



OPENROCKET SIMULATION

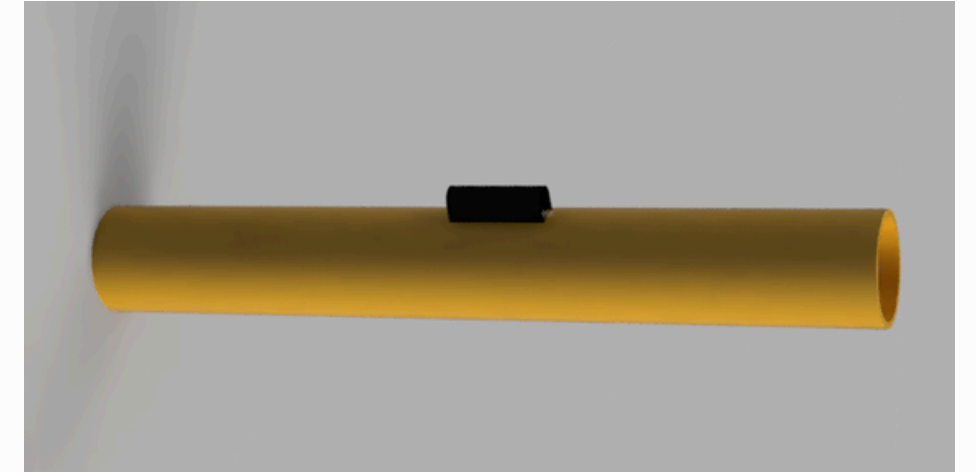
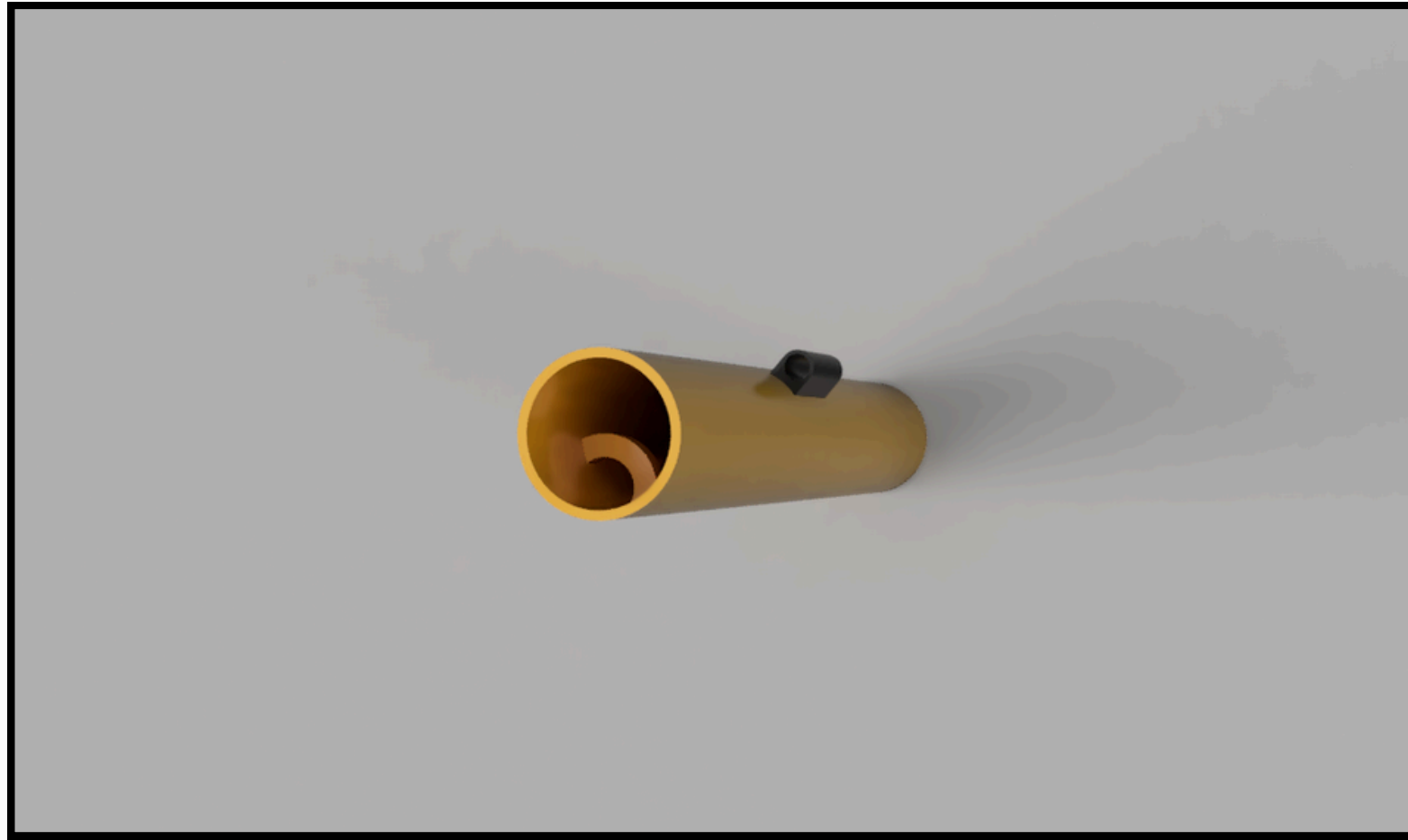


01

- Elliptical form
- NOSE FINS
- Shoulder

DESIGNING THE NOSE CONE





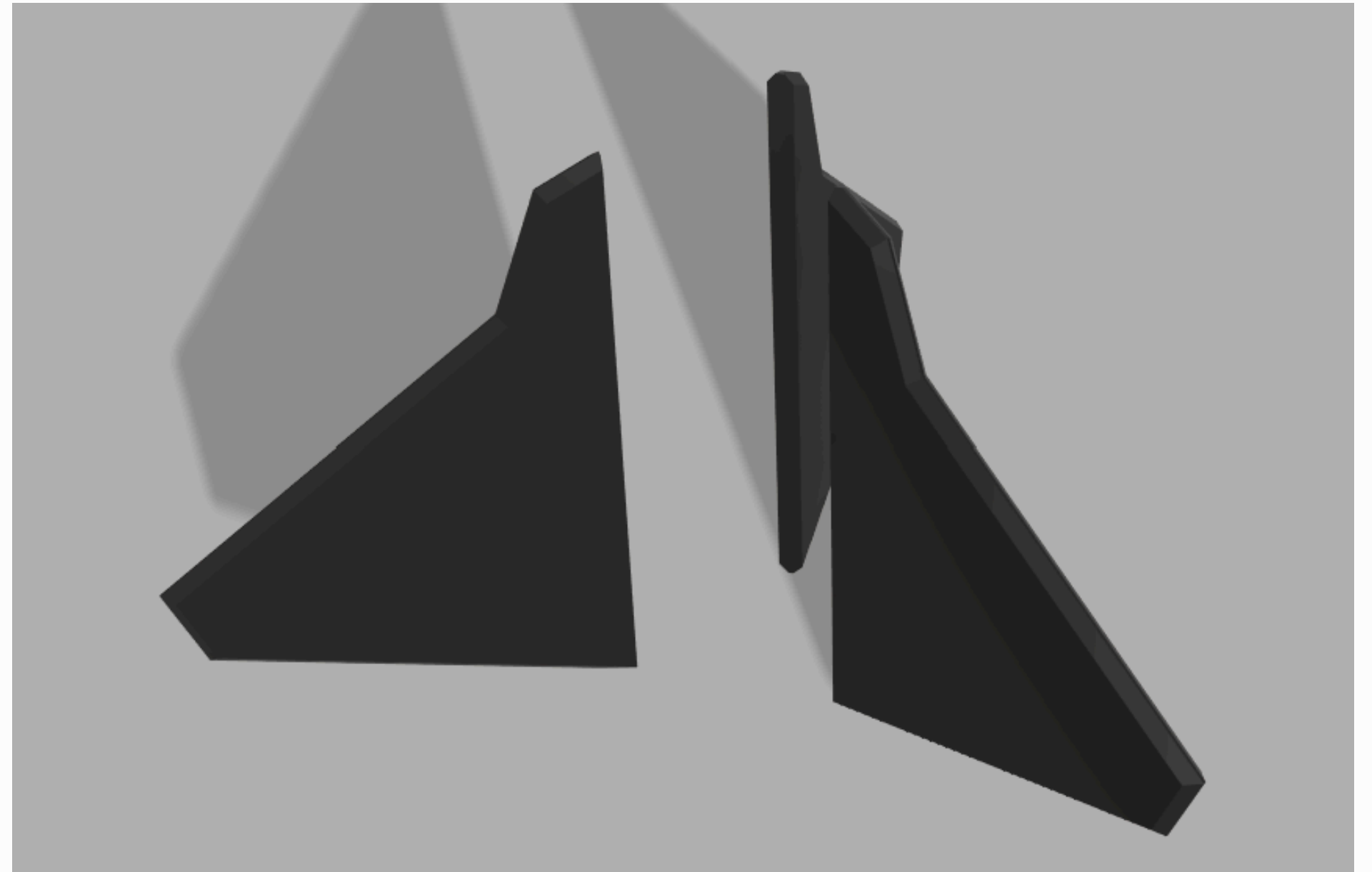
01

- Ring - cause of major issues
- 1 Launch lug
- Narrow

DESIGNING THE BODY TUBE

01

- Freeform
- 3 fins
- EPIC shape



DESIGNING THE FINS

OUR LAUNCH

FLAWLESS.

Upon Launching:

The rocket launched vertically, with some angular direction initially

Recovery

We found the recovery attempt left more to be desired

- parachute did not deploy (not taking enough account in getting the parachute to actually be ejected after attaching to bulkhead)
 - Shock cord remained attached (YAY)
- Everything is still in tact
-





REFLECTION



- Many aspects could have been improved upon
- **Placement of the shock cord ring:**
 - By attaching our shock cord at the ring closer to the top, the parachute was given little space and could not properly eject
 - **IMPROVEMENT:** lower ring
 - longer shock cord to connect both nose cone and body tube

Positives:
LOCKHEED APPROVED
(job security)

